2. Group the same factors in pairs.
3. If each prime factor is grouped then the number is a perfect square.
4. If any factor is left from pairing then the number is not a perfect square.

Example: Test whether the following numbers are perfect squares or not.

(i) 5625 (ii) 4563

(i) Solution:

Prime factorization = $\underline{5 \times 5} \times \underline{5 \times 5} \times \underline{3 \times 3}$

Since each prime factor is in pairs,

So, 5625 is a perfect square.

5	5625
5	1125
5	225
5	45
3	9
	3

(ii) Solution:

Prime factorization is : $\underline{3 \times 3} \times 3 \times \underline{13 \times 13}$

As, 3 is left from pairing,

So, 4563 is not a perfect square.

3	4563
3	1521
3	507
13	169
13	13
	1

5.1.3 Properties of perfect square of a number.

Identify and apply some interesting properties of numbers and perfect squares.

(i) The square of even number is even.

We know that the numbers exactly divisible by 2 are called even numbers. Below are the perfect squares of some even numbers.

Even Number	Square	Perfect square
2	$2^2 = 2 \times 2$	4
4	$4^2 = 4 \times 4$	16
6	$6^2 = 6 \times 6$	36
8	$8^2 = 8 \times 8$	64
10	$10^2 = 10 \times 10$	100

Comparing the even numbers and their perfect squares, we conclude that squares of all even numbers are also even.

(ii) The square of an odd number is odd.

We know that odd numbers are not exactly divisible by 2. Let us identify perfect squares of some odd numbers.

Odd Number	Square	Perfect square
1	$1^2 = 1 \times 1$	1
3	$3^2 = 3 \times 3$	9
5	$5^2 = 5 \times 5$	25
7	$7^2 = 7 \times 7$	49
9	$9^2 = 9 \times 9$	81

From the above table it is clear that the squares of all odd numbers are also odd.

(iii) The square of a proper fraction is less than itself:

We know that a fraction is called proper fraction if its denominator is greater that its numerator. Let us learn the process of squaring and comparing a proper fraction by the following examples.

Example. Find the square of $\frac{3}{4}$ and compare with itself.

Solution: Let us find square of $\frac{3}{4}$.

$$\left(\frac{3}{4}\right)^2 = \left(\frac{3}{4}\right) \times \left(\frac{3}{4}\right) = \frac{3 \times 3}{4 \times 4} = \frac{9}{16}$$

Comparing equivalent fractions

Given number is: $\frac{3}{4} = \frac{3 \times 4}{4 \times 4} = \frac{12}{16}$, and $\left(\frac{3}{4}\right)^2 = \frac{9}{16}$.

Here $\frac{9}{16} < \frac{12}{16}$, so square of the given number is less than the given number.

i.e. $\frac{9}{16} < \frac{3}{4}$ or $\left(\frac{3}{4}\right)^2 < \frac{3}{4}$

Therefore the square of a fraction is less than itself.

Similarly: $\left(\frac{2}{9}\right)^2 = \left(\frac{2}{9}\right) \times \left(\frac{2}{9}\right) = \frac{2 \times 2}{9 \times 9} = \frac{4}{81}$ and $\frac{4}{81} < \frac{2}{9}$

$\left(\frac{1}{7}\right)^2 = \left(\frac{1}{7}\right) \times \left(\frac{1}{7}\right) = \frac{1 \times 1}{7 \times 7} = \frac{1}{49}$ and $\frac{1}{49} < \frac{1}{7}$

(iv) The square of a decimal less than 1 is smaller than the given decimal.

Consider the following examples.

Example: Find the square of 0.2 and compare with itself

Solution: Square: $(0.2)^2 = (0.2) \times (0.2) = \frac{2}{10} \times \frac{2}{10} = \frac{4}{100} = 0.04$

It is observed that: $0.04 < 0.2$

or $(0.2)^2 < 0.2$

Therefore the square of a decimal less than one is always smaller than the given decimal.

Similarly: $(0.03)^2 = (0.03) \times (0.03) = \frac{3}{100} \times \frac{3}{100} = \frac{9}{10000} = 0.0009$

So, $0.0009 < 0.03$

or $(0.03)^2 < 0.03$

Also $(0.12)^2 = (0.12) \times (0.12) = \frac{12}{100} \times \frac{12}{100} = \frac{144}{10000} = 0.0144$

So, $0.0144 < 0.12$

or $(0.12)^2 < 0.12$

EXERCISE 5.1

A. Find squares of the following.

(1) 11 (2) 19 (3) 25 (4) 50

(5) 66 (6) 78 (7) 100 (8) 500

B. Test whether each of the following is a perfect square or not.

(1) 81 (2) 95 (3) 121 (4) 169

(5) 224 (6) 3969 (7) 3872 (8) 9845

C. Separate the perfect square of even and odd numbers.

(i) 9 (ii) 25 (iii) 196 (iv) 441 (v) 2704

(vi) 3600 (vii) 9216 (viii) 9801

D. Find the squares of the following proper fractions and compare with themselves.

(1) $\frac{1}{5}$ (2) $\frac{3}{7}$ (3) $\frac{8}{9}$ (4) $\frac{5}{6}$ (5) $\frac{2}{3}$

E. Find the squares of the following decimal numbers and compare with themselves.

(1) 0.1 (2) 0.5 (3) 0.07 (4) 0.11 (5) 0.15

5.2 SQUARE ROOT

5.2.1 Define square root of a natural number and recognize its notation.

Square root of a number is a numerical value which when multiplied by itself, gives the original number. It is a useful tool to solve the mathematical equations related to our daily life. Square root was introduced by an ancient Egyptian who used it in construction and other activities.

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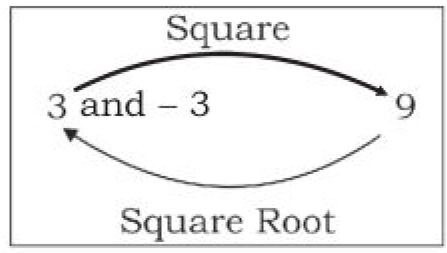
We know how to find a perfect square. The process of finding square root is just opposite to it.

As $3^2 = 3 \times 3 = 9$ also $(-3)^2 = (-3) \times (-3) = 9$, we read it as 'the square of 3 and (– 3) is 9'. For square root, we read it as '3 and (– 3) are the square roots of 9.'

Therefore every natural number has two square roots. One is negative and the other is positive, we take only positive root at this stage.



The symbol of radical sign '$\sqrt{\ }$' is used to show square root of a number. The square root of 9 can be written as $\sqrt{9}$ where 9 is called the radicand.

In general square root is used to solve the equation of type $x^2 - y = 0$ or $x^2 = y$ when y is called the square of x and x is called the square root of y.

5.2.2 Find square root by division method and factorization method of natural number, fraction and decimals.

There are two ways to find square root of a number i.e. by factorization method and by division method.

(i) Square root by factorization method:

Example 1: Find square root of the following by factorization method.

(i) 256 (ii) 3969

(i) Solution: First of all we find the prime factorization of 256.

2	256
2	128
2	64
2	32
2	16
2	8
2	4
2	2
	1

$256 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$

Now $\sqrt{256} = \sqrt{\underline{2 \times 2} \times \underline{2 \times 2} \times \underline{2 \times 2} \times \underline{2 \times 2}}$

$= 2 \times 2 \times 2 \times 2$ (Pairing factors)

$= 16$

Therefore $\sqrt{256} = 16$

Verification: $16 \times 16 = 256$

(ii) Solution: First we find the prime factorization of 3969.

3	3969
3	1323
3	441
3	147
7	49
7	7
	1

$3969 = 3 \times 3 \times 3 \times 3 \times 7 \times 7$

Now $\sqrt{3969} = \sqrt{3 \times 3 \times 3 \times 3 \times 7 \times 7}$

or $\sqrt{3969} = 3 \times 3 \times 7$ (Pairing factors)

$= 63$

Therefore, $\sqrt{3969} = 63$

Verification: $63 \times 63 = 3969$

Example 2:

Find square root of the following fractions by factorization method.

(i) $\frac{64}{81}$ (ii) $2\frac{113}{256}$

(i) Solution: We find the square root of the numerator and denominator separately as:

Finding the prime factorization of 64 and 81, we get:

Numerator: $64 = \underline{2 \times 2} \times \underline{2 \times 2} \times \underline{2 \times 2}$

Denominator: $81 = \underline{3 \times 3} \times \underline{3 \times 3}$

$$\sqrt{\frac{64}{81}} = \frac{\sqrt{64}}{\sqrt{81}} = \frac{\sqrt{\underline{2 \times 2} \times \underline{2 \times 2} \times \underline{2 \times 2}}}{\sqrt{\underline{3 \times 3} \times \underline{3 \times 3}}}$$

$$= \frac{2 \times 2 \times 2}{3 \times 3} = \frac{8}{9}$$

Numerator

2	64
2	32
2	16
2	8
2	4
2	2
	1

Denominator

3	81
3	27
3	9
3	3
	1

(ii) Solution: $2\frac{113}{256} = \frac{625}{256}$ (by changing the mixed fraction into improper fraction.)

Finding prime factorization of 625 and 256, we get:

Numerator: $625 = \underline{5 \times 5} \times \underline{5 \times 5} = 5^2 \times 5^2$

Denominator: $256 = \underline{2 \times 2} \times \underline{2 \times 2} \times \underline{2 \times 2} \times \underline{2 \times 2}$

Numerator	
5	625
5	125
5	25
5	5
	1

Denominator	
2	256
2	128
2	64
2	32
2	16
2	8
2	4
2	2
	1

So, $\sqrt{2\frac{113}{256}} = \sqrt{\frac{625}{256}}$

or $\sqrt{2\frac{113}{256}} = \frac{\sqrt{625}}{\sqrt{256}} = \frac{\sqrt{\underline{5 \times 5} \times \underline{5 \times 5}}}{\sqrt{\underline{2 \times 2} \times \underline{2 \times 2} \times \underline{2 \times 2} \times \underline{2 \times 2}}}$

$= \frac{5 \times 5}{2 \times 2 \times 2 \times 2}$

or $\sqrt{2\frac{113}{256}} = \frac{25}{16} = 1\frac{9}{16}$

Example 3: Find square root of the following decimals by factorization method.

(i) 1.44 (ii) 0.81

(i) Solution: $1.44 = \frac{144}{100}$ (By converting decimal into common fraction).

Numerator	
2	144
2	72
2	36
2	18
3	9
3	3
	1

Denominator	
2	100
2	50
5	25
5	5
	1

Now $\sqrt{1.44} = \sqrt{\frac{144}{100}} = \frac{\sqrt{144}}{\sqrt{100}}$

or $\sqrt{1.44} = \frac{\sqrt{\underline{2 \times 2} \times \underline{2 \times 2} \times \underline{3 \times 3}}}{\sqrt{\underline{2 \times 2} \times \underline{5 \times 5}}} = \frac{2 \times 2 \times 3}{2 \times 5} = \frac{12}{10}$

or $\sqrt{1.44} = 1.2$ (By converting into decimal).

(ii) Solution: $0.81 = \frac{81}{100}$

Numerator	
3	81
3	27
3	9
3	3
	1

Denominator	
2	100
2	50
5	25
5	5
	1

$\sqrt{0.81} = \sqrt{\frac{81}{100}} = \frac{\sqrt{81}}{\sqrt{100}}$

or $\sqrt{0.81} = \frac{\sqrt{\underline{3 \times 3} \times \underline{3 \times 3}}}{\sqrt{\underline{2 \times 2} \times \underline{5 \times 5}}} = \frac{3 \times 3}{2 \times 5}$

or $\sqrt{0.81} = \frac{9}{10} = 0.9$

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(ii) Square root by division method:

This method is very useful to find square root of numbers. First of all the digits of the radicand must be paired. Let us learn to find the square root of perfect square by division method of natural numbers, fractions and decimals with the help of following examples.

Example 1: Find square root of the following natural numbers by division method.

(i) 256 (ii) 1024

(i) Solution:

Step 1: Start pairing the digits by putting a bar over each pair from ones place and so on (In case of odd number of digits the digit on extreme left will be left alone and considered as a pair itself as $\hat{2}\overline{56}$).

Step 2: Consider the left most pair, i.e. '2' find the largest number whose square be either 2 or very near but less than 2 which is '1'.

Step 3: Subtract square of $1(1^2 = 1 \times 1 = 1)$ from the pair i.e, $2 - 1 = 1$ and bring down the next pair which is '56'.

Step 4: Add divisor '1' to itself and make it the digit at left most place of the new divisor.

Step 5: Again find the largest number whose multiplication at ones with the divisor must be equal to or less than that of new dividend (156).

i.e. $\boxed{26 \times 6 = 156}$

When remainder becomes zero and all digits of given number are used then process stops.

$$
\begin{array}{r|r}
 & 16 \\
\hline
1 & \hat{2}\overline{56} \\
+1 & -1 \\
\hline
26 & 156 \\
+6 & -156 \\
\hline
32 & 0
\end{array}
$$

Step 6: Quotient is the required square root in division method which can be verified by squaring.

Hence, $\sqrt{256} = 16$

(ii) Solution:

$$
\begin{array}{r|r}
 & 32 \\
\hline
3 & \overline{10}\,\overline{24} \\
+3 & -9 \\
\hline
62 & 124 \\
+2 & -124 \\
\hline
64 & 0
\end{array}
$$

So, $\sqrt{1024} = 32$

Example 2: Find square root of $1\frac{1089}{1936}$ by division method.

Solution:

As $1\frac{1089}{1936} = \frac{3025}{1936}$

Numerator: $\overline{30}\ \overline{25}$

$$\begin{array}{r|l} & 55 \\ \hline 5 & \overline{30}\ \overline{25} \\ +5 & -25 \\ \hline 105 & 525 \\ +5 & -525 \\ \hline 110 & 0 \end{array}$$

Denominator: $\overline{19}\ \overline{36}$

$$\begin{array}{r|l} & 44 \\ \hline 4 & \overline{19}\ \overline{36} \\ +4 & -16 \\ \hline 84 & 336 \\ +4 & -336 \\ \hline 88 & 0 \end{array}$$

Hence, $\sqrt{1\frac{1084}{1936}} = \sqrt{\frac{3025}{1936}} = \frac{\sqrt{3025}}{\sqrt{1936}} = \frac{\cancel{55}^{5}}{\cancel{44}_{4}} = \frac{5}{4} = 1\frac{1}{4}$

So, $\sqrt{1\frac{1089}{1936}} = 1\frac{1}{4}$

Example 3: Find square root of the following decimals by division method.

Solution: (i) 10.24 (ii) 249.64 (iii) 0.0441

(i)
$$\begin{array}{r|l} & 3.2 \\ \hline 3 & \overline{10}.\overline{24} \\ +3 & -9 \\ \hline 62 & 124 \\ +2 & -124 \\ \hline 64 & 0 \end{array}$$

(ii)
$$\begin{array}{r|l} & 15.8 \\ \hline 1 & \hat{2}\overline{49}.\overline{64} \\ +1 & -1 \\ \hline 25 & 149 \\ +5 & -125 \\ \hline 308 & 2464 \\ +8 & -2464 \\ \hline 316 & 0 \end{array}$$

(iii)
$$\begin{array}{r|l} & 0.21 \\ \hline 2 & 0.\overline{04}\ \overline{41} \\ 2 & -4 \\ \hline 41 & 041 \\ +1 & -41 \\ \hline 42 & 0 \end{array}$$

Hence $\sqrt{10.24} = 3.2$ So, $\sqrt{249.64} = 15.8$ So, $\sqrt{0.0441} = 0.21$

Note: Pairing in decimals, start from either sides of the decimal point, '0' can be placed to complete the pair of the decimal part.

EXERCISE 5.2

A. Find square root of each of the following.

(1)	25	(2)	$(4)^2$	(3)	81	(4)	$(36)^2$
(5)	a^2	(6)	y^2	(7)	$(49)^2$	(8)	64

B. Find square root by factorization and division methods.

(1)	676	(2)	169	(3)	484	(4)	961
(5)	4900	(6)	1089	(7)	1600	(8)	2304
(9)	3136	(10)	1681	(11)	1369	(12)	2025

C. Find the square root of the following fractions by factorization and division method.

(1)	$\frac{144}{225}$	(2)	$\frac{324}{441}$	(3)	$\frac{1225}{9801}$	(4)	$1\frac{25}{144}$
(5)	$1\frac{48}{121}$	(6)	$3\frac{13}{36}$	(7)	$\frac{961}{1681}$	(8)	$\frac{1024}{1225}$
(9)	$2\frac{217}{576}$	(10)	$5\frac{55}{169}$	(11)	$1\frac{63}{81}$	(12)	$3\frac{325}{900}$

D. Find the square root of the following decimals by factorization and division methods.

(1)	3.24	(2)	4.41	(3)	5.29	(4)	7.29
(5)	6.25	(6)	37.21	(7)	7.84	(8)	10.24
(9)	30.25	(10)	100.00	(11)	33.64	(12)	34.81

5.2.3 Solve real life problems involving square roots.

In our daily life, there are so many problems, in which we use square root (directly or indirectly) to solve them. Let us` consider the following examples.

Unit 5 SQUARE ROOT OF POSITIVE NUMBER

Example: Students of class VII collected money for picnic. Each student contributed as much rupees as the number of students present. If Rs. 4761 is the total collection, find the amount paid by each.

Solution: Total amount = 4761 rupees

In this case amount paid by each student = number of students

= square root of 4761

(Division Method)

	69
6	$\overline{47}\,\overline{61}$
+ 6	– 36
129	1161
+ 9	– 1161
138	0

3	4761
3	1587
23	529
23	23
	1

(Prime factorization Method)

$$\sqrt{4761} = \sqrt{3 \times 3 \times 23 \times 23}$$
$$= 3 \times 23$$
$$= 69$$

Hence, each student contributed Rs 69.

Example 2: There are 529 students in a school. All the students are standing in rows. There are as many rows, as the number of students standing in each row. Find the number of rows.

Solution:

Total number of students = 529

There are as many rows, as the number of students in each row.

Therefore we can find the number of rows, by finding the square root of the total number of students.

Here $\sqrt{529}$ = 23

So, there are 23 rows and in each row there are 23 students.

	23
2	$\hat{5}\,\overline{29}$
+ 2	– 4
43	129
+ 3	129
46	0

Example 3: There are 1225 plants arranged in as many rows, as there are plants in each row. Find the number of plants in each row.

Solution: Total number of plants = 1225 plants.

Number of rows = Number of plants in each row.

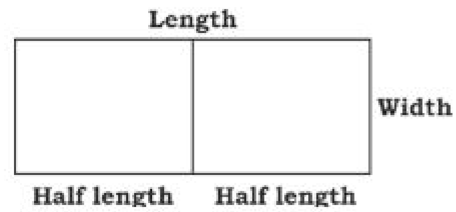
Therefore, the number of plants in each row = $\sqrt{1225}$ = 35.

Thus, there are 35 rows of plants and in each row there are 35 plants.

	35
3	$\overline{12}\,\overline{25}$
+3	– 9
65	325
+5	– 325
70	0

Example 4: The length of a rectangular ground is twice its width. If its area is 96800 square metres, find length of the rectangular ground.

Solution:



If the ground is divided in two equal parts, each will be a square.

The area of

each square $= \dfrac{96800}{2}$

$= 48400$ square metres.

Length of each square $= \sqrt{\text{area of square}}$

$= \sqrt{48400}$

$= 220$ m

Hence length of the ground $= 2 \times$ length of square

$= 2 \times 220 = 440$ m.

	220
2	$\hat{4}\ \overline{84}\ \overline{00}$
+ 2	− 4
42	84
+2	− 84
440	000
+ 0	− 000
440	0

EXERCISE 5.3

1. 784 chairs are arranged in an examination hall in such a way that the number of chairs in each row is the same as the number of rows. Find the number of rows. Also find the number of chairs in each row.
2. The area of a square field is 240.25 square metres. Calculate the length of its side.
3. The length of a room is three times of its width and the area is 720.75 square are metres. Find its length and width.
4. There are 2116 packets of biscuits packed in a carton. There are as many rows of biscuits as there are biscuits in each row. Find the number of rows of biscuits.
5. School function is to be attended by 1681 students. Find the number of chairs required for seating arrangement in such a way that equal number of students sit in a row as well as in column.

6. There are 625 trucks. All these trucks are standing in rows. There are as many rows, as there are trucks in each row. Find the number of rows and the number of trucks in each row.

7. A drill teacher arranged 1024 students to stand in rows. There are as many rows as there are students in each row. How many number of rows are there and what is the number of students in each row?

8. The length of a rectangular field is thrice as its width. If the area of the field is 995.9052 square metres. Find its width and length.

9. Find the length of boundary of a square shaped school whose area is 3387.24 square metres.

10. In a sugar factory 15,198 bags of sugar are to be arranged into the shape of a solid square. But it is found that 69 bags are in excess. How many bags are arranged in a row?

REVIEW EXERCISE 5

1. Give short answers of the following.

(i) What are the different ways to find the square root of a number?

(ii) What is a perfect square number? Give examples.

(iii) Describe any two properties of perfect square of a number. Give it examples.

2. Fill in the blanks.

(i) The square of 31 is _________.

(ii) The square root of 49 is _________.

(iii) 1, 4, 9, 16,.. are called _________ numbers.

(iv) 144 is the _________ of 12.

(v) $\sqrt{\frac{25}{36}}$ = _________ (vi) Square of 1 is _________

3. Write "T" for true and "F" for false.

(i) Prime factors of 235 are 2, 3 and 5.

(ii) $19^2 = 361$ (iii) The square of an odd number is even.

(iv) $\frac{25}{49} > \frac{5}{7}$ (v) $(4.5)^2 > 4.5$ (vi) $0.7 < (0.7)^2$

4. Choose the correct answer.

(i) If $p = q^2$ then q is called _________ of p.

(a) square root (b) square

(c) prime factor (d) radicand

(ii) The numbers whose squares are equal to their square roots are _________.

(a) 0 and 2 (b) 1 and 3 (c) 0 and 1 (d) 2 and 3

(iii) The perfect square number is _________.

(a) 14 (b) 15 (c) 16 (d) 17

5. Find square root of the following:

(i) 729 (ii) 7056 (iii) 26569 (iv) 42025

(v) $5\frac{71}{121}$ (vi) $\frac{2116}{9604}$ (vii) $1\frac{984}{14641}$ (viii) 0.0256

(ix) 344569 (x) $\frac{6400}{9801}$ (xi) $5\frac{41}{64}$ (xii) 131.1025

6. The area of a square shaped park is 12100 square metres. Find the length of its side.

7. A floor of a square shaped room has an area of 144 square metres. Find the length of the floor.

8. The area of a rectangular field is 7688 square metres. Find the width when length of the field is twice of its width.

9. The length of a window is thrice its width. Find the length and width of the window when its area is 18.75 square metres.

10. Najeeb spent Rs 900 in the month of June in such a way that he spent as many rupees in a day as the number of days in the month. How many rupees did he spend in a day?

SUMMARY

- The numerical value which we get by multiplication of a number with itself is called its square.
- Perfect square is a positive number which is the square of any number.
- Prime factors of a perfect square are always in the pairs.
- The square of an even number is even and square of an odd number is odd.
- The square of a proper fraction is less than itself.
- The square of a decimal less than 1 is smaller than the given decimal.
- Square root is a number which when multiplied by itself, gives the original number.
- The process of finding the square root is the reverse operation of squaring a number.
- "$\sqrt{\ }$" is used to show square root. It is called radical sign.
- The number inside radical sign is called radicand.
- We can find the square root of a mixed fraction, by converting it into an improper fraction.
- Let x, y be any two numbers then:

  (i) $\sqrt{x \times y} = \sqrt{x} \times \sqrt{y}$ (ii) $\sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}}$
- In the equation $x^2 = y$, y is called the square of x and x is called the square root of y.
- There are two ways to find square root of a number; factorization method and division method.
- Square root of a decimal can be found by changing it into a common fraction.